THE LIFTING PROPERTY (EQUATION-SMOOTH) IMPLIES SMOOTHNESS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2344-2356 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 2358-2402, the complete author proof (43 body lines).

Editorial note. Navigation stubs: 10 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 20 of queue_artin.json (pb=2113, stmt 2344-2356, proof 2358-2402).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 03YK: sources/stacks-artin/pinned-original-source.tex, source lines 2402–2402. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07WI: sources/stacks-artin/pinned-original-source.tex, source lines 2402–2402. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07XS: sources/stacks-artin/pinned-original-source.tex, source lines 2268–2275. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07XV: sources/stacks-artin/pinned-original-source.tex, source lines 2315–2321. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07XU: sources/stacks-artin/pinned-original-source.tex, source lines 2302–2307. Cited by the retained statement or proof but not part of

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the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07XM: sources/stacks-artin/pinned-original-source.tex, source lines 1693–1706. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07XN: sources/stacks-artin/pinned-original-source.tex, source lines 1733–1742. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07XW: sources/stacks-artin/pinned-original-source.tex, source lines 2332–2338. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02HX: sources/stacks-artin/pinned-original-source.tex, source lines 2402–2402. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02J4: sources/stacks-artin/pinned-original-source.tex, source lines 2402–2402. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a locally Noetherian scheme. Let $f : \mathcal{X} \to \mathcal{Y}$ be a 1-morphism of categories fibred in groupoids over $(Sch/S)_{fppf}$. Assume*

1. *f is representable by algebraic spaces,*
2. *f satisfies (07XS),*
3. *$\mathcal{X} \to (Sch/S)_{fppf}$ is limit preserving on objects, and*
4. *$\mathcal{Y}$ is limit preserving.*

Then f is smooth.

Proof. The key ingredient of the proof is More on Morphisms, Lemma 02HX which (almost) says that a morphism of schemes of finite type over S satisfying (07XS) is a smooth morphism. The other arguments of the proof are essentially bookkeeping.

Let V be a scheme over S and let y be an object of $\mathcal{Y}$ over V. Let Z be an algebraic space representing the 2-fibre product $\mathcal{Z} = \mathcal{X} \times_{f,\mathcal{X},y} (Sch/V)_{fppf}$. We have to show that the projection morphism $Z \to V$ is smooth, see Algebraic Stacks, Definition 03YK. In fact, it suffices to do this when V is an affine scheme locally of finite presentation over S, see Criteria for Representability, Lemma 07WI. Then $(Sch/V)_{fppf}$ is limit preserving by Lemma 07XN. Hence $Z \to S$ is locally of finite presentation by Lemmas 07XM and 07XN. Choose a scheme W and a surjective étale morphism $W \to Z$. Then W is locally of finite presentation over S.

Since f satisfies (07XS) we see that so does $\mathcal{Z} \to (Sch/V)_{fppf}$, see Lemma 07XV. Next, we see that $(Sch/W)_{fppf} \to \mathcal{Z}$ satisfies (07XS) by Lemma 07XW. Thus the composition

$$(Sch/W)_{fppf} \to \mathcal{Z} \to (Sch/V)_{fppf}$$

satisfies (07XS) by Lemma 07XU. More on Morphisms, Lemma 02HX shows that the composition $W \to Z \to V$ is smooth at every finite type point w_0 of W. Since the smooth locus is open we conclude that $W \to V$ is a smooth morphism of

schemes by Morphisms, Lemma 02J4. Thus we conclude that $Z \to V$ is a smooth morphism of algebraic spaces by definition. □